

GlaucoScan — RE-GAN Analysis Report

Fundus Image Preprocessing & Clinical Feature Extraction

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60.0%

High Risk Significant structural features consistent with glaucoma.
Urgent ophthalmology referral recommended.

Preprocessing Results

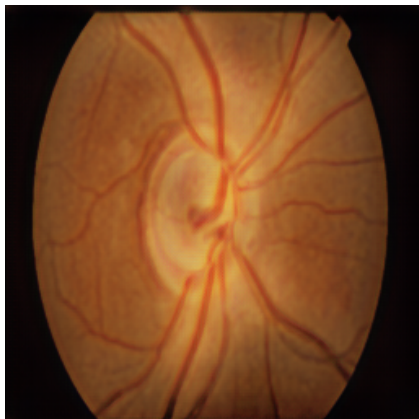
The fundus image was processed through a 5-stage classical pipeline followed by RE-GAN deep learning enhancement. The three images below show the progression from raw input to fully enhanced output.



Original (Raw input)



Classical Pipeline (Stages 1–5)



RE-GAN Enhanced (Deep learning)

Classical pipeline: 53.0 ms

RE-GAN enhancement: 4217.3 ms

Total: 4270.3 ms

Extracted Clinical Features

The following five biomarkers were extracted from the RE-GAN-enhanced image using the two-stage optic disc localisation algorithm. All features are grounded in published ophthalmological literature.

Feature	Value	Clinical Range	Interpretation	Weight
Vertical CDR	0.657	< 0.65 normal	Elevated (>0.65 threshold)	50%
Horizontal CDR	0.723	< 0.65 normal	Elevated (>0.65 threshold)	—
Rim-Disc Area Ratio	0.497	> 0.50 normal	Borderline rim loss	25%
ISNT Score	2 / 3	3 = fully normal	Minor violation	15%
Vessel Density	1.17%	> 10% normal	Reduced vessel density	10%

Optic Disc & Cup Measurements

Measurement	Value	Measurement	Value
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Disc Height (px)	170.4	Cup Height (px)	112.0
Disc Width (px)	77.1	Cup Width (px)	55.7
Disc Centre X	129	Image Sharpness	24.43
Disc Centre Y	121	Enhancement	RE-GAN (U-Net+CBAM)

Glaucoma Risk Score Breakdown

The overall glaucoma probability of 60.0% is computed using a weighted clinical scoring formula based on published ophthalmological thresholds. Each feature contributes independently to the final score.

Feature	Raw Score	Weight	Contribution
Vertical CDR	70%	50%	35.0%
Rim-Disc Area Ratio	50%	25%	12.5%
ISNT Score	33%	15%	5.0%
Vessel Density	75%	10%	7.5%
TOTAL		100%	60.0%

Preprocessing Pipeline Used

Stage	Operation	Purpose	Status
1	Green Channel Extraction	Maximise disc/vessel contrast	✓ Applied
2	Gaussian BG Normalisation	Remove non-uniform illumination (Novel)	✓ Applied
3	CLAHE Enhancement	Boost local contrast	✓ Applied
4	Bilateral Denoising	Edge-preserving noise removal	✓ Applied
5	Gamma Correction	Brightness adjustment	✓ Applied
6	RE-GAN Enhancement	Deep learning enhancement (U-Net+CBAM)	✓ Applied

Important Notice

This report is generated by an automated research system (GlaucoScan RE-GAN) developed as part of a PhD research programme. It is intended for research and educational purposes only and does NOT constitute a medical diagnosis. All findings must be confirmed by a qualified ophthalmologist before any clinical decision is made. If you are experiencing vision problems, please consult a healthcare professional immediately.